

Prime Quadruple Resonators and Logarithmic Spectral Fluctuations

Thomas Hoffbauer

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Abstract

We introduce a family of observables associated with prime quadruples $(p, p+2, p+6, p+8)$ arranged in mod-12 families (E, A, B, C) . For each quadruple we define a resonance defect based on the geometric mean of three primes.

After removing the asymptotic expansion of this defect we analyze the residual fluctuations in the logarithmic variable $u = \log x$. Numerical evidence suggests that these fluctuations exhibit oscillatory components compatible with frequencies associated with the imaginary parts of the nontrivial zeros of the Riemann zeta function.

This motivates a heuristic spectral interpretation of prime quadruples as local arithmetic resonators and suggests a discrete operator model in the spirit of the Hilbert–Pólya program.

1 Introduction

The connection between the distribution of prime numbers and the nontrivial zeros of the Riemann zeta function

$$\zeta(s)$$

is encoded in the explicit formulas of analytic number theory. These formulas express fluctuations in prime counting functions as oscillatory contributions involving the imaginary parts γ of the zeros

$$\rho = \frac{1}{2} + i\gamma.$$

In this work we investigate whether similar oscillatory signatures appear in observables derived from prime quadruples

$$(p, p+2, p+6, p+8).$$

These quadruples form two mod-12 families

$$(A, B, C, E), \quad (C, E, A, B)$$

which will play the role of two chiral sectors of the system.

2 Prime Quadruple Defects

Consider a prime quadruple

$$(p, p+2, p+6, p+8).$$

Define

$$(a, b, c, e) = (p, p+2, p+6, p+8).$$

We introduce the resonance defect

$$\Delta_{ABCE}(a) = e - (abc)^{1/3} - \frac{16}{3}.$$

For the second ordering

$$(c, e, a, b) = (p, p+2, p+6, p+8)$$

we define

$$\Delta_{CEAB}(c) = e - (abc)^{1/3} + 4.$$

These quantities measure the deviation between the largest prime of the quadruple and the geometric mean of the preceding three primes.

3 Asymptotic Expansion

For large primes the defect admits an expansion

$$\Delta(x) = \frac{K_1}{x} + \frac{K_2}{x^2} + \frac{K_3}{x^3} + O(x^{-4}).$$

Explicit calculation yields

$$\Delta_{ABCE}(a) = \frac{28}{9a} - \frac{832}{81a^2} + \frac{10288}{243a^3} + O(a^{-4})$$

and

$$\Delta_{CEAB}(c) = \frac{52}{9c} - \frac{1624}{81c^2} + \frac{23008}{243c^3} + O(c^{-4}).$$

Removing these smooth contributions isolates the fluctuating component.

4 Logarithmic Spectral Analysis

Define

$$u = \log x$$

and the detrended signal

$$y(u) = \Delta(x) - \Delta_{\text{asym}}(x).$$

If the fluctuations are influenced by the zeta spectrum we expect an expansion

$$y(u) \approx \sum_{\gamma} A_{\gamma} \cos(\gamma u + \phi_{\gamma}).$$

We therefore project the signal onto the basis

$$\cos(\gamma \log x), \quad \sin(\gamma \log x)$$

using the first several hundred imaginary parts γ of the zeta zeros.

5 Numerical Results

The projections produce amplitudes

$$\mathcal{A}_{\gamma} = \sqrt{A_{\gamma}^2 + B_{\gamma}^2}.$$

Empirically we observe that certain frequencies appear repeatedly in both families of prime quadruples.

Figures illustrate

- the detrended defect as a function of $\log x$
- the reconstruction using the strongest spectral modes
- the amplitude spectrum across γ

6 Operator Heuristic

The oscillatory structure suggests interpreting prime quadruples as elements of a discrete state space

$$\mathcal{H} = \text{span}\{|q\rangle\}$$

where each $|q\rangle$ corresponds to a quadruple.

Define a diagonal observable

$$H_0|q\rangle = \Delta(q)|q\rangle.$$

Couplings between nearby quadruples define an interaction operator T .

The total operator

$$H = H_0 + T$$

or in chiral form

$$D = \begin{pmatrix} 0 & T^* \\ T & 0 \end{pmatrix}$$

may produce spectral statistics related to the imaginary parts of the zeta zeros.

7 Geometric Embedding

To visualize the internal structure we embed quadruples into an eight-dimensional space

$$(\log p_1, \log p_2, \log p_3, \log p_4, \phi_E, \phi_A, \phi_B, \phi_C)$$

where the phases encode the mod-12 prime families.

Projection of this space reveals preferred directions and possible chiral separation between the two quadruple types.

8 Discussion

The proposed observables define a new arithmetic object associated with prime quadruples.

While the observed oscillations do not constitute a proof of any spectral interpretation, they motivate further investigation of discrete operator models connecting prime configurations with zeta spectra.

Future work includes

- extending the numerical range of quadruples
- studying stability of spectral peaks
- constructing explicit discrete operators on quadruple graphs

9 Conclusion

Prime quadruples equipped with geometric-mean defects provide a new framework for studying oscillatory phenomena in prime distributions.

The logarithmic spectral analysis suggests a possible resonance structure compatible with frequencies related to the imaginary parts of the Riemann zeta zeros.

References

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